

POSITION CONTROL SERVO SYSTEM

Consider the servo system shown in Figure. In control engineering, a servo system is an automatic device that uses error-sensing negative feedback to correct the action of the mechanism. The operation of the system is explained below.

The error measuring device is the pair of potentiometers which convert the input and output position in to proportional voltages. The desired output position is given as the input to the first potentiometer. The electric potential of the arm is proportional to the angular position r of the arm. The output shaft position determines the angular position c of the arm of the output potentiometer. The difference between the input angular position r and the output angular position c is the error signal e

$$e = r - c$$

e_r is the input voltage corresponding to the position r and e_c is the output voltage corresponding to output position c .

$$e_r = K_0 r \quad \text{and} \quad e_c = K_0 c$$

where K_0 is a proportionality constant.

Therefore, the error voltage is

$$e_v = e_r - e_c$$

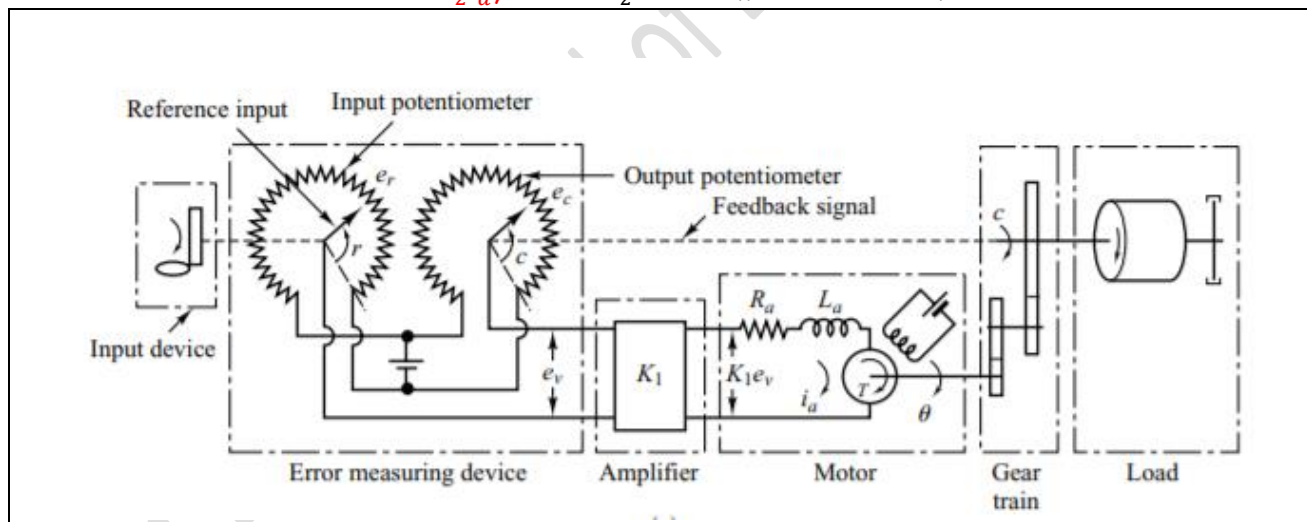
This error voltage is applied to the amplifier with gain K_1 . The output voltage of the amplifier

$$K_1 e_v = e_a$$

is applied to the armature circuit of the dc motor. A fixed voltage is applied to the field winding. When there is an error, the motor rotates so that the error becomes zero.

If the field current is constant, the torque developed by the motor is proportional to the armature current i_a

$$T = K_2 i_a, \text{ where } K_2 \text{ is the motor constant.}$$



When the armature is rotating, a voltage proportional to the product of the flux and angular velocity is induced in the armature. For a constant flux (constant field voltage), the induced voltage (or the back emf)

$$e_b = K_3 \frac{d\theta}{dt}, \text{ where } K_3 \text{ is the back emf constant of the motor and } \theta \text{ is the angular displacement of the motor shaft.}$$

The speed of the armature-controlled dc servomotor is determined by the armature voltage e_a . The differential equation for the armature circuit is

$$e_a = L_a \frac{di_a}{dt} + R_a i_a + e_b$$

$$e_a = L_a \frac{di_a}{dt} + R_a i_a + K_3 \frac{d\theta}{dt} \dots \dots \dots (a)$$

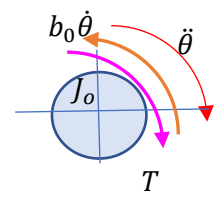
The torque developed by the motor is $T = K_2 i_a$

The torque equilibrium (Eqn of rotation)

$$J_0 \ddot{\theta} = T - b_0 \dot{\theta}$$

$$J_0 \ddot{\theta} + b_0 \dot{\theta} = T = K_2 i_a \dots \dots \dots (b)$$

Where J_0 is the mass moment of inertia of the motor, load, and gear train with respect to the motor shaft axis and b_0 is the viscous friction coefficient of the combination of the motor, load, and gear train referred to the motor axis.



Taking Laplace Transforms of equations (a) and (b)

$$E_a(s) = L_a s I_a(s) + R_a I_a(s) + K_3 s \theta(s) \dots \dots \dots (c)$$

$$J_0 s^2 \theta(s) + b_0 s \theta(s) = K_2 I_a(s) \dots \dots \dots (d)$$

From (d) $I_a(s) = \frac{(J_0 s^2 + b_0 s) \theta(s)}{K_2}$ which when substituted in to equation (c) gives

$$E_a(s) = (L_a s + R_a) \frac{(J_0 s^2 + b_0 s) \theta(s)}{K_2} + K_3 s \theta(s) \dots \dots \dots (e)$$

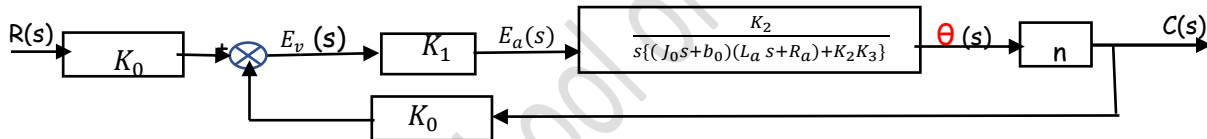
$$\frac{\theta(s)}{E_a(s)} = \frac{K_2}{(J_0 s^2 + b_0 s)(L_a s + R_a) + K_2 K_3 s} = \frac{K_2}{s\{(J_0 s + b_0)(L_a s + R_a) + K_2 K_3\}} \dots \dots \dots (f)$$

If the gear ratio of the gear train is n (i.e., for each revolution of the motor shaft, the output shaft rotates n times), then

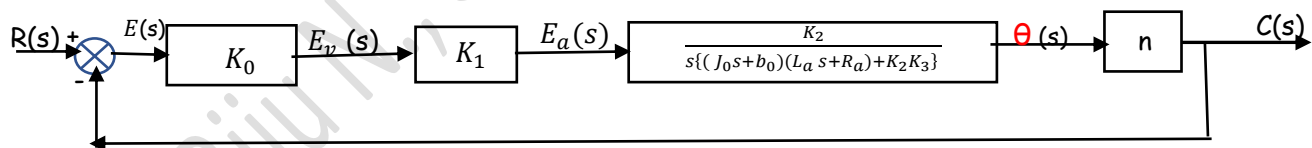
$$C(s) = n \theta(s) \dots \dots \dots (g)$$

We have $e_v = e_r - e_c = K_0 r - K_0 c$. i.e.,

$$E_v(s) = E_r(s) - E_c(s) = K_0 \{R(s) - C(s)\} = K_0 E(s) \dots \dots \dots (h)$$

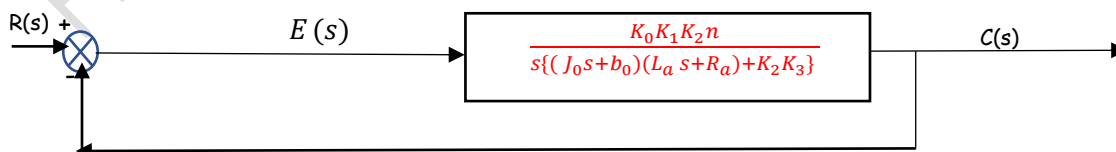


If both the potentiometer constants are the same, we can combine them as shown below and make a unity feed back system.



If we look at the forward path, the series blocks can be combined with the transfer function

$$G(s) = \frac{C(s)}{E(s)} = \frac{K_0 K_1 K_2 n}{s\{(J_0 s + b_0)(L_a s + R_a) + K_2 K_3\}}$$



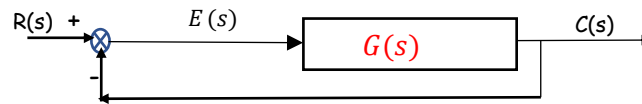
If L_a is very small

$$G(s) = \frac{K_0 K_1 K_2 n}{s\{(J_0 s + b_0)R_a + K_2 K_3\}} = \frac{\frac{K_0 K_1 K_2 n}{R_a}}{s\{(J_0 s + b_0) + \frac{K_2 K_3}{R_a}\}} = \frac{\frac{K_0 K_1 K_2 n}{R_a}}{J_0 s^2 + \left(b_0 + \frac{K_2 K_3}{R_a}\right)s}$$

If we multiply J_0 and $b_0 + \frac{K_2 K_3}{R_a}$ by $\frac{1}{n^2}$, then the inertia and damping are referred to the output shaft axis.

$$J = \frac{J_0}{n^2}, \quad B = \frac{b_0 + \frac{K_2 K_3}{R_a}}{n^2}, \quad K = \frac{K_0 K_1 K_2 n}{R_a}$$

$$G(s) = \frac{K}{Js^2 + Bs}$$



The overall transfer function,

$$T(s) = \frac{G}{1 + GH} = \frac{\frac{K}{Js^2 + Bs}}{1 + \frac{K}{Js^2 + Bs}} = \frac{K}{Js^2 + Bs + K}$$

Reference

1. Modern Control Engineering : Katsuhiko Ogatta, Pearson Education, LPE